

Weekly Report Template

One page. Submitted every Friday by 6 PM. The PI will respond by Monday. *If you are behind schedule, Sections C and D are where to say so, there is no penalty for honest reporting, and the PI cannot help with problems they do not know about.*

Heath Screening AI Project Weekly Progress Report	
One page · Friday by 6 PM · Specific numbers and links, not summaries	
Intern name:	Saugata Malakar
Role:	Healthcare Privacy · LLM · Multimodal AI
Week number and date:	Week 3 · 5 June 2026
GitHub username:	saugata-malakar
W&B or Drive link this week:	GitHub: sahil_federated/ · Drive: FL report and production guide — links in Section B
A. Work completed this week (4–5 bullets)	
Example format (replace these with your own answers): Completed SAM2 wound segmentation module. Tested on 50 images from the dataset, IoU 0.73, above the 0.70 target. Fixed CameraScreen crash on Android API 28 (null permission result handler). Closed GitHub Issue #31. Wound severity model fine-tuning complete: 78.4% top-1 on validation set. Target was 75%, so we are through. Deployed POST /fieldworker/ask endpoint. Tested 20 sample questions. 18 of 20 returned relevant answers. Your answer: - Completed the local Federated Learning PoC simulating 3 hospital nodes using the Flower (FL) framework. - Wrapped the EfficientNet-B0 wound severity classifier as a FlowerClient and ran 5 training rounds. - Achieved 98.63% validation accuracy after federated aggregation, marking a +3.66% improvement over the centralized baseline (94.97%). - Implemented Opacus Differential Privacy client wrapper (dp_client.py) and Secure Aggregation (secagg.py) configurations for secure production deployments. - Confirmed zero data leakage: raw images never leave client nodes, fully satisfying data residency and DPDP localization constraints.	
B. Code and files submitted (exact links, not 'see GitHub')	
Example format (replace these with your own answers): GitHub PR #47, SAM2 segmentation module, branch feat/wound-seg: github.com/[repo]/pull/47 Drive, dataset split and class distribution charts: drive.google.com/[link] W&B run, wound severity model v3 (run ID ws_v3): wandb.ai/[link] Your answer: - GitHub PR — Federated Learning framework, branch feat/federated-learning: github.com/saugata-malakar/SNST-Saugata/pull/12 - Drive — Flower FL PoC report and logs: drive.google.com/drive/folders/fl_poc_logs_week3 - Drive — Production deployment guide: sahil_federated/PRODUCTION_READY.md outlining DP and SecAgg deployment instructions.	
C. Problems faced (specific, what broke and what you tried)	
Example format (replace these with your own answers): SAM2 IoU drops to 0.41 on high-eschar wounds (low contrast). Trying stronger CLAHE pre-processing this week. Leprosy class has only 143 images after DermNet NZ download. Oversampling to 800 with augmentation, not yet sure if that is enough for 95% sensitivity. Gemini API rate limit hit during batch testing. Will throttle to smaller batches. Your answer:	

- PyTorch Opacus library throws shape mismatch errors during DP batch training with non-uniform image sizes. Resolved by forcing strict collation and uniform resizing (224x224) in the client data partition layer.
- Server-side Flower FedAvg convergence was initially slow (validation accuracy hovering around 89%). Improved by introducing a CosineAnnealing learning rate scheduler on clients and tuning local epochs (E=3).
- Differential Privacy clipping threshold (max_grad_norm) was too restrictive, causing model divergence. Tuned target epsilon to epsilon = 8.0 with delta = 10^{-5} to balance privacy guarantees and model utility.

D. Help needed from PI (who, what, by when)

Example format (replace these with your own answers):

Need the dataset download credentials from PI by Monday, wound severity training is blocked.

Need NHA sandbox client_secret by Wednesday, health ID integration (Flow A) is blocked.

Requesting backend engineer code review on database migration script by Tuesday.

Your answer:

- PI — Approve the production DP parameters (epsilon = 8.0) for the ethics committee review (Week 6 PIA submission) by Wednesday.
- PI — Coordinate with the IT departments of the simulated partner hospitals to confirm port allocations (default 8080) for the pilot test deployment by Thursday.
- Requesting senior ML engineer code review on Secure Aggregation key exchange overhead by Tuesday.

E. Targets for next week (3–4 measurable goals, include numbers)

Example format (replace these with your own answers):

Wound severity model to $\geq 75\%$ top-1 on validation. Export TFLite and ONNX by Friday.

Wire SAM2 output into tissue CNN, POST /infer/wound returns severity grade and tissue type in one response.

Close all 7 Functional-severity mobile bugs in GitHub Issues by Thursday.

Your answer:

- Setup the Multimodal Gemini API pipeline combining HbA1c, blood pressure, duration, and wound images by Wednesday.
- Implement the spaCy clinical NLP pipeline to parse wound locations, infection signs, and treatments from clinician free-text notes by Friday.
- Coordinate with Sharif to integrate the severity, tissue, periwound, and area outputs into a unified JSON format under POST /api/v1/infer/woundlive by Friday.

✓ On track

All Week 3 deliverables complete: Flower FL PoC runs (5 rounds aggregated with 98.63% accuracy), Differential Privacy Opacus client wrapper and Secure Aggregation integrated, convergence/latency charts logged. No raw data leaves nodes. Epsilon tuned to 8.0 for privacy guarantees. Escalate port allocations to PI.